

BOROMO, Burkina Faso

WMO: 655160

Lat: 11.750N

Lon: 2.933W

Elev: 889

StdP: 14.23

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.4	63.4	16.6	13.1	85.9	20.8	16.1	85.8	12.2	86.8	11.6	86.1	0.3	40	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.0	104.7	70.0	103.2	70.1	101.8	70.3	80.1	91.5	79.5	90.9	79.0	90.4	4.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	147.9	84.6	76.8	144.5	84.2	76.2	141.5	83.8	44.4	91.7	43.8	90.8	43.2	90.6	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	9.8	8.1	DB	56.7	108.6	1.9	1.1	55.4	109.4	54.3	110.0	53.2	110.7	51.9	111.5	
			WB	44.4	81.5	1.4	1.0	43.4	82.2	42.6	82.8	41.8	83.4	40.8	84.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	83.8	77.9	83.5	89.5	91.3	89.4	85.4	81.7	80.0	81.3	84.1	83.2	78.9	(6)
(7)		DBStd	5.27	3.49	4.41	3.48	3.55	3.77	3.67	3.06	2.28	2.51	2.35	2.55	3.08	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	12346	865	939	1224	1238	1220	1063	981	931	939	1056	995	895	(10)
(11)		CDD65	6871	400	519	759	788	755	613	516	466	489	591	545	430	(11)
(12)		CDH74	88031	4678	6884	11671	12329	11180	7970	5436	4241	4824	7035	6744	5039	(12)
(13)		CDH80	49092	2460	4075	7569	8113	6919	4175	2222	1392	1927	3569	3944	2727	(13)
(14)	Wind	WSAvg	2.5	2.6	3.0	3.0	3.3	3.6	3.0	2.3	1.9	1.5	1.7	1.5	2.3	(14)
(15)	Precipitation	PrecAvg	35.0	0.0	0.0	0.3	1.5	3.1	4.7	7.3	9.9	6.3	1.7	0.2	0.0	(15)
(16)		PrecMax	44.7	1.1	0.5	2.3	6.7	7.6	10.2	12.6	17.4	11.0	5.6	1.8	1.2	(16)
(17)		PrecMin	21.4	0.0	0.0	0.0	0.0	0.0	1.9	2.5	3.7	2.9	0.0	0.0	0.0	(17)
(18)		PrecStd	5.7	0.2	0.1	0.4	1.3	1.7	1.6	2.4	3.0	2.1	1.4	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.4	104.8	106.2	106.9	104.9	101.0	96.2	91.7	94.2	98.9	100.2	98.7	(19)
(20)			MCWB	63.8	66.6	67.5	71.5	74.2	77.4	77.6	79.0	79.0	73.1	67.8	64.8	(20)
(21)		2%	DB	96.1	101.9	104.3	105.0	102.6	98.5	93.2	89.9	92.4	96.9	98.8	96.2	(21)
(22)			MCWB	61.5	65.7	67.5	71.5	75.7	77.7	77.8	78.4	78.9	74.6	67.6	63.5	(22)
(23)		5%	DB	93.7	99.8	103.0	103.3	100.7	96.5	91.2	88.4	90.8	95.2	97.6	94.3	(23)
(24)			MCWB	60.6	64.7	67.5	71.8	76.2	77.5	77.6	77.9	78.7	75.4	67.6	62.9	(24)
(25)		10%	DB	91.5	97.3	101.6	101.5	98.8	94.4	89.4	86.9	89.0	93.5	96.2	92.3	(25)
(26)			MCWB	59.9	63.6	67.4	72.4	76.5	77.3	77.3	77.5	78.3	76.1	67.6	62.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.9	74.3	77.3	79.2	80.8	80.1	80.0	80.1	81.0	80.7	77.2	73.2	(27)
(28)			MCDWB	80.1	93.3	96.8	96.2	95.3	93.5	89.8	88.4	90.1	90.6	90.9	89.2	(28)
(29)		2%	WB	64.0	71.8	75.6	78.2	79.8	79.3	79.1	79.2	79.9	79.5	75.5	67.4	(29)
(30)			MCDWB	92.5	91.2	94.9	95.3	94.3	92.7	89.1	87.3	89.3	90.0	89.1	90.6	(30)
(31)		5%	WB	62.2	69.0	74.5	77.4	78.9	78.7	78.4	78.5	79.2	78.6	74.2	64.7	(31)
(32)			MCDWB	90.6	91.4	93.4	94.5	93.5	92.1	88.1	86.3	88.5	89.6	88.2	90.6	(32)
(33)		10%	WB	60.7	65.6	73.2	76.6	78.2	78.1	77.8	77.9	78.5	77.9	72.8	63.1	(33)
(34)			MCDWB	89.1	95.0	92.0	93.6	92.6	90.8	87.0	85.2	87.4	88.8	87.1	89.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	27.4	26.5	23.8	20.0	18.3	16.6	14.5	12.9	15.1	19.7	28.6	28.4	(35)
(36)			MCDBR	30.7	29.5	25.8	22.5	20.5	19.2	17.0	15.2	17.7	22.5	29.7	30.4	(36)
(37)			MCWBR	10.6	10.9	9.8	7.6	5.3	5.0	5.3	6.0	6.9	6.2	8.4	10.0	(37)
(38)		5% WB	MCDWB	29.6	24.8	20.4	19.1	19.0	17.5	15.6	14.2	16.7	18.9	23.1	28.6	(38)
(39)			MCWBR	11.0	11.6	9.0	6.6	5.7	5.5	5.9	6.2	7.1	6.7	7.1	10.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.569	0.637	0.752	0.714	0.664	0.618	0.553	0.525	0.527	0.589	0.552	0.525	(40)	
(41)		taud	1.802	1.664	1.491	1.637	1.704	1.814	2.012	2.089	2.061	1.884	1.871	1.908	(41)	
(42)		Ebn at Noon	235	225	203	210	217	224	240	251	252	234	238	243	(42)	
(43)		Edh at Noon	68	81	97	83	77	68	56	52	54	64	63	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1794	1874	1937	1949	1921	1817	1715	1643	1748	1866	1798	1756	(44)	
(45)		RadStd	91	94	89	77	72	71	67	62	58	50	51	61	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page